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Interactive Road traffic crash analysis and predictive
modelling for Queensland

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Abstract:

This study aims to analyze and predict traffic accident data in Queensland by using machine learning techniques to identify high-risk areas and predict the severity of accidents. The data comes from traffic accident records in Queensland from 2001 to 2023, combined with auxiliary data such as road roughness to improve the accuracy of prediction. The study uses a variety of models such as Random Forest, Support Vector Machine (SVM) and Density Clustering (DBSCAN) to deeply explore the environment and factors of accident occurrence through data fusion and feature extraction. The results show that the addition of road condition data can improve the prediction effect of severe accidents, while the DBSCAN clustering model effectively locates accident hotspots with dense populations and high traffic volume. The data-driven framework provided by this study can support Queensland's traffic safety strategy, thereby optimizing resource allocation and reducing the probability of accidents. Future research can consider introducing dynamic risk assessment in the time dimension and more features to further improve the accuracy and application value of prediction.

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1.Introduction

Traffic accidents are a major global public safety and public health challenge, causing millions of casualties and billions of dollars in economic losses every year. According to statistics from the World Health Organization, traffic accidents are one of the leading causes of death and disability worldwide, especially in low- and middle-income countries (World Health Organization, 2018). In this context, the prediction of traffic accidents and the identification of high-risk areas have become key topics in traffic management and safety strategy formulation (Chen, Zhang, & Li, 2019). However, current research still has significant shortcomings in the integration of data sets, model selection, and prediction accuracy (Li, Wu, & Zhao, 2021). This article uses random forest, support vector machine and DBSCAN to conduct multi-dimensional comprehensive analysis, aiming to provide a more efficient method for traffic accident severity prediction and high-risk area identification, thereby providing decision-makers with more targeted traffic safety Improvement plan.

As traffic conditions become more complex and road conditions vary across regions, it becomes more difficult to control traffic accidents (Elvik & Vaa, 2004). Likewise for drivers, various factors such as weather, road surface roughness, and the driver's own physical condition may have direct and indirect effects on the occurrence of traffic accidents (Chang & Chen, 2018). According to the World Health Organization (WHO), approximately 1.35 million people died in road traffic accidents worldwide in 2016, with an average of 18.2 deaths per 100,000 people (WHO, 2018). This highlights the ongoing and urgent need to address the factors that lead to road accidents to control the loss of life and property caused by traffic crashes (Peden et al., 2004).

The topic of this research is "Interactive Road Traffic Crash Analysis and Predictive Modelling for Queensland." Through this research topic, the main purpose is to predict the Queensland state traffic crash severity and to identify the high traffic crash risk area. Through these two purposes, that can help Queensland government to know the which area are easily happen traffic crash and has an estimation for the crash severity. Based on that to adjust the traffic strategy, balance police power distribution and highlight the road situation. Then can effectively reduce the traffic crash happen probability.

The research methodology involves analyzing crash data from various perspectives to gain insights into the circumstances surrounding each crash. For example, examining driver behavior, such as whether the

driver wore a helmet or consumed alcohol, can reveal potential risk factors. While some factors may directly cause accidents, others might serve as underlying contributors that increase the likelihood of crashes occurring. By thoroughly analyzing this data, the study aims to accurately predict the time, location, and severity of traffic accidents. Consequently, the primary objective of this project is to forecast crash severity and identify high-risk regions. After obtaining these results, the next step will be to perform time-series analysis to pinpoint the times of highest risk for each state.

Following this introductory section, the research paper is organized into five main parts: Literature Review, Research Problem, ethical consideration, Objectives, Modelling, and Conclusion. The Literature Review will discuss current methods used for analyzing crash severity and identifying high-risk regions or suburbs, as well as the challenges associated with these analyses. The Research Problem section will present a problem formulation based on insights from the literature review. Ethical consideration part will discuss Data security and privacy risk. In the Objectives section, the specific goals this project aims to achieve will be detailed. The Modelling section will then outline the machine learning models and techniques employed for this research, providing a step-by-step explanation of how each model contributes to the study. Finally, the Conclusion section will conclude all the section of this research

2.Relative work and challenge.

That is not a new problem in every state, so there are also some professional articles that relevant with the traffic crash analysis. This literature review aims to provide a comprehensive understanding of the existing research on traffic accident prediction models and the impact of road conditions on traffic safety (Zhao & Zheng, 2020). By examining prior studies, this review will identify gaps in the current literature and establish the foundation for the research presented in this study (Lee & Mannering, 2002).

Traffic crash analysis and predictive modeling have been widely studied across the globe, employing a range of methodologies from traditional statistical techniques to advanced machine learning approaches (Wang & Yao, 2021). Researchers have been exploring various methods to improve the accuracy of traffic crash predictions, understand the factors influencing crash severity, and develop effective prevention strategies. The choice of method for these

studies has varied significantly, reflecting the evolving landscape of data analysis techniques in the field of traffic safety research.

Dr. Khaled J. Assi was study for conducted by focused on developing a hybrid system that integrates multiple analytical techniques to predict traffic crash severity. In his research, Dr. Assi employed a combination of Principal Component Analysis (PCA) with Multilayer Perceptron (MLP) neural networks and Support Vector Machines (SVMs) to enhance the prediction accuracy. The dataset used in this study comprised a total of 74,909 crashes that occurred in Victoria, Australia, over a period of five years from 2014 to 2019. The Multilayer Perceptron neural network, a subset of artificial neural networks, is known for its capability to mimic human cognitive processes and handle complex nonlinear relationships. This model leverages both computational and human-like intelligence to capture intricate patterns within data, thereby providing improved generalization performance.

On the other hand, the Support Vector Machine (SVM) is a powerful machine learning algorithm widely utilized for both classification and regression problems. SVM is particularly effective in high-dimensional data analysis due to its ability to maximize the margin between different classes, thus achieving high generalization performance (Burges, 1998). In Dr. Assi's study, SVMs were employed alongside MLPs to analyze the dataset and predict crash severity, taking advantage of their complementary strengths in handling different types of data patterns and complexities.

Before implementing the primary predictive models, Dr. Assi also conducted preliminary tests to assess the dataset's suitability for multivariate analysis. Two statistical tests were employed: the Kaiser-Meyer-Olkin (KMO) test and Bartlett's test of sphericity. These tests were crucial in evaluating the dataset's multicollinearity and ensuring that the variables were appropriate for Principal Component Analysis (PCA). The KMO test measures sampling adequacy to determine if the dataset is suitable for factor analysis, while Bartlett's test checks whether the correlation matrix of the dataset is an identity matrix, indicating that the variables are unrelated. By performing these tests, Dr. Assi ensured the robustness and reliability of the subsequent predictive modeling process.

However, despite the strengths of this hybrid approach, there were certain limitations in Dr. Assi's study. One significant issue was that the study only considered crashes involving two vehicles or single-vehicle incidents. As a result, the findings and the predictive model's applicability might not extend to multi-vehicle crashes (involving more than two vehicles), which can have different dynamics and severity factors. This limitation suggests that future research should incorporate a more comprehensive range of crash types to develop more universally applicable predictive models.

Another significant study in this domain was conducted by Xiao Wen, Yuan Chang Xie, Liming Jiang, Ziyuan Pu, and Tingjian Ge, who focused on the application of various machine learning methods for traffic crash severity modeling. Unlike Dr. Assi's hybrid model, this research employed a broader spectrum of machine learning algorithms, including Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), Decision Trees, Bayesian Methods, Random Forests, K-Nearest Neighbors (KNN), and Association Rule Mining. By utilizing this diverse range of techniques, the researchers aimed to compare and contrast the strengths and weaknesses of each method in predicting traffic crash severity.

In their study, the researchers identified several critical challenges that could impact the results of traffic crash severity modeling. One of the most notable issues discussed was the problem of data imbalance and underreporting. Traffic crashes, particularly those with less severe outcomes, are often underreported, leading to an imbalanced dataset where certain types of crashes are overrepresented while others are underrepresented. This imbalance can skew the model's predictions, resulting in a biased understanding of crash severity dynamics. Addressing these data-related issues is essential for improving the reliability and accuracy of predictive models in traffic crash analysis.

Overall, these studies illustrate the complexity and multi-faceted nature of traffic crash severity modeling. The combination of different machine learning techniques and the careful consideration of data quality issues can provide more accurate and comprehensive insights into the factors contributing to traffic crashes. Future research in this area could benefit from integrating more diverse data sources, employing advanced data augmentation techniques to

address imbalance, and developing models that can accommodate a wider range of crash types and conditions.

3.Dataset overview:

To achieve these research objectives, this study will use two primary data sets: Queensland traffic accident data and road system condition-roughness data.

1. Queensland traffic accident data: This dataset consists of six sub-datasets, comprehensively covering the overview of reported road traffic accidents in Queensland between January 1, 2001 and May 31, 2023:

- The first sub-dataset: Road Accident Locations - provides statistical information on accident locations and accident characteristics within Queensland. Analysis of this sub-dataset will help identify high-risk accident areas and help develop targeted interventions to address Improve road safety.
- The second sub-dataset: road casualties - records in detail the characteristics of casualties in traffic accidents, including accident type, victim category, injury severity and other information. This dataset allows us to further explore the impact of different types of accidents on casualties, providing key input to accident severity prediction models.
- The third sub-dataset: Driver demographics - summarizes the information of drivers involved in accidents, including age, gender, driving experience and other information. Analysis combined with driver information will reveal the participation rates and risk factors of different groups in traffic accidents.
- The fourth sub-dataset: seat belt and helmet use - contains seat belt use by vehicle occupants, as well as helmet use by motorcycle and cyclists and its association with accident outcomes. This helps evaluate the effectiveness of safety precautions in reducing accident casualties.
- The fifth sub-dataset: Vehicle Type - records the types of vehicles involved in traffic accidents, which provides a basis for analyzing the roles and risks of different types of vehicles in traffic accidents.
- The sixth sub-dataset: Road accident factors - includes the involvement of alcohol, fatigue, vehicle failure and other factors in traffic accidents, helping to identify potential causes of accidents and provide guidance for intervention measures.

2. Road System Conditions - Roughness Data: This dataset provides road condition information through the "Road System Conditions - Roughness 100m" sub-dataset. Specifically, this sub-dataset measures and classifies the roughness condition of the road in units of 100 meters, using NAASRA roughness counts for averaging and classification. This data helps analyze the impact of road conditions on traffic accidents, thereby revealing the relationship between road maintenance and safety, and providing additional environmental variables for traffic accident risk prediction models.

Comprehensive analysis of these two datasets will help build traffic accident severity prediction models, identify high-risk areas, and provide data-based support for future road safety strategies.

Based on the dataset and our objectives, for this research the select the crash location dataset and the roughness 100m dataset be used to analysis.

4.Ethical consideration:

All the data used in this study are from the database of the Queensland Government's official open data portal website. All the data used are used for traffic crash analysis to prevent information leakage. The data may contain some sensitive traffic information such as the location and time of the relevant traffic crash, as well as the severity of the traffic crash and subsequent processing. In this study, the data are stored on the local computer and are only used for crash severity forecast and high-risk region identification. The model may have bias in predicting high-risk areas for traffic accidents. Some geographical areas may be systematically marked as high-risk due to historical data, which may lead to biased resource allocation. The model will not have a discriminatory impact on specific areas or groups, for example, it will not produce unfair high-risk predictions for low-income communities due to data distribution issues. The research results may have an impact on travel in some areas, such as transportation and tourism. If the research results are used by the government, the results are still highly regional and there will be limitations in the use of the model. For example, the prediction results in some areas may be biased. The data is only used for research and improvement of public interests such as traffic safety, and will not be used for other commercial or non-public purposes.

5. Research problem:

In this study, we aim to use machine learning techniques to analyze and predict the severity of traffic accidents and identify high-risk areas in Queensland. This study is based on an existing database containing comprehensive traffic accident data from 2001 to 2023, which is our primary dataset. To improve prediction accuracy and gain deeper insights into accident patterns, we integrate additional data on road conditions (particularly road roughness) from the Queensland Government's official open data portal.

The `crash_location` table in our primary database provides detailed spatial and temporal information for each traffic accident, while the `road_roughness_100m` dataset (our auxiliary data) provides road surface quality data at 100-meter intervals in Queensland. By combining these datasets, we aim to build a more nuanced understanding of the factors that contribute to accident severity.

Using this integrated dataset, our research focuses on two main objectives:

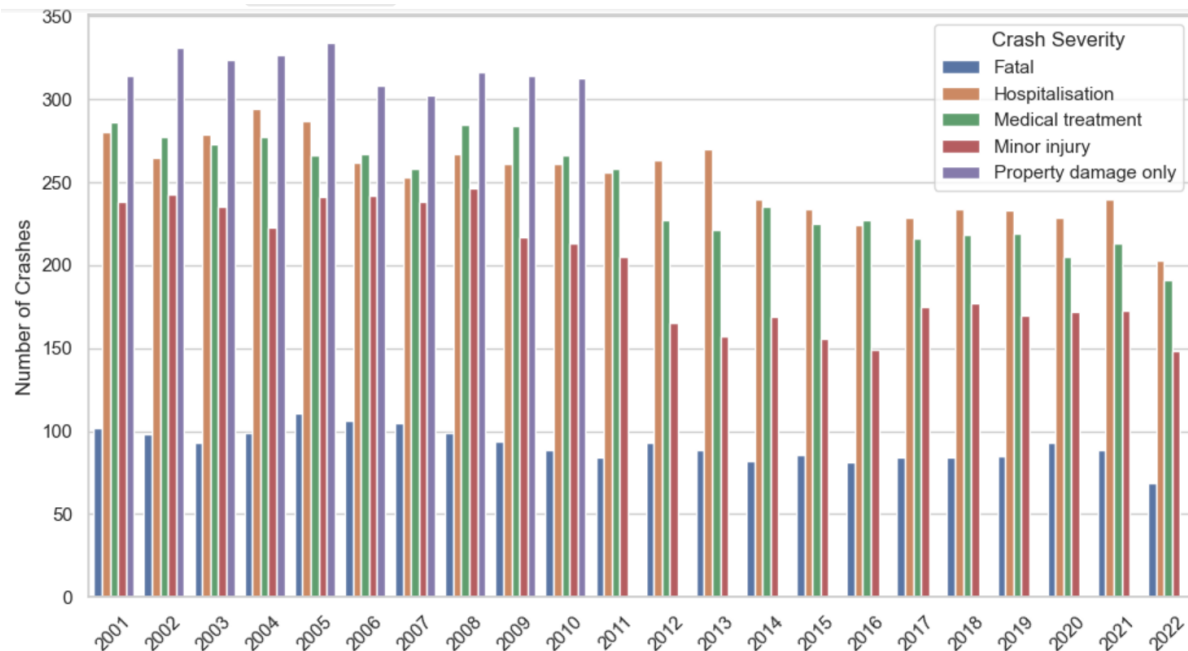
Traffic accident severity prediction: Develop predictive models based on historical accident patterns, road roughness, and other contextual factors to predict the severity of future accidents. These models will provide insights into the likelihood of severe accidents under different road and environmental conditions.

High-risk area identification: Identify areas with a higher likelihood of a serious crash. By analysing the spatial correlation between road conditions and crash data, we seek to identify specific areas in Queensland that are at higher risk of a serious crash. This information is critical for policymakers and the road safety sector to priorities interventions and allocate resources.

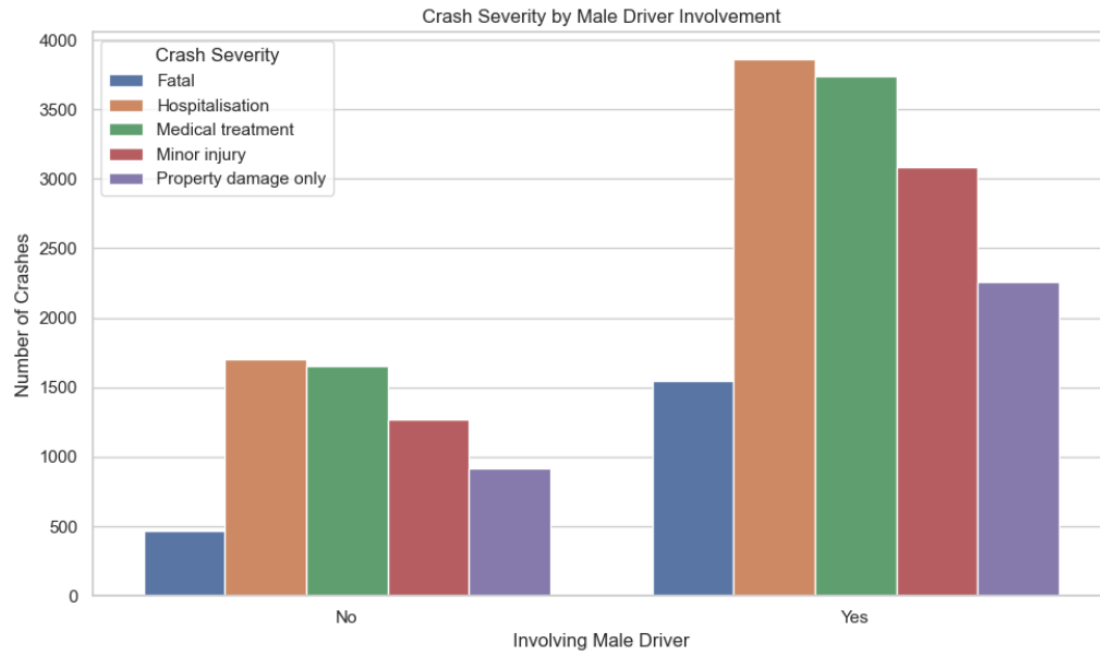
The results of this study may help strengthen road safety measures and help reduce the incidence and severity of crashes in high-risk areas.

6. EDA analysis

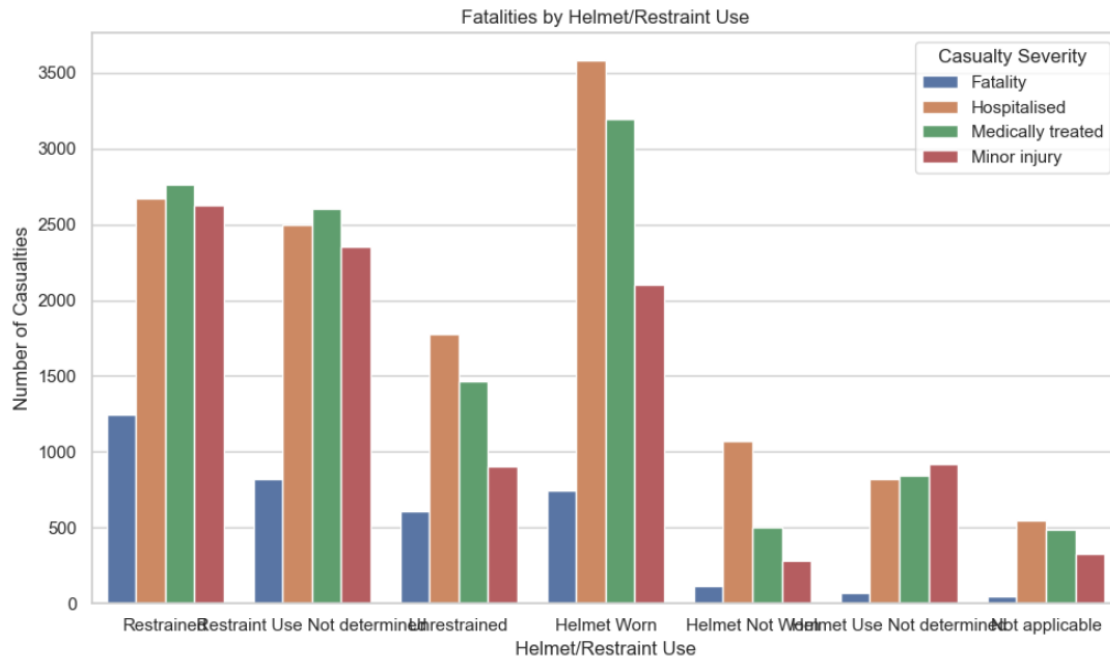
Data analysis:



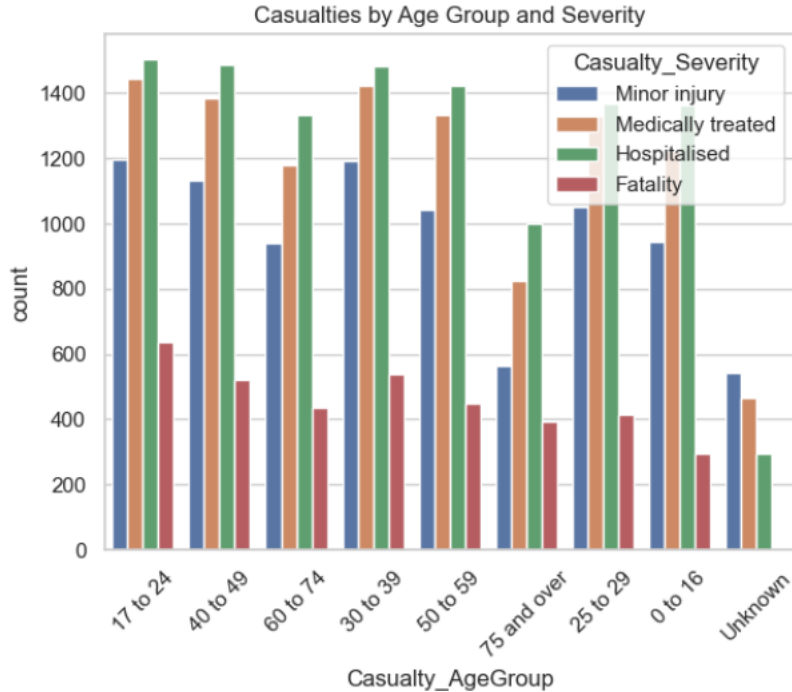
This chart shows the changes in the number of road accidents between 2001 and 2022, classified by accident severity. The five types of accidents are: fatal accidents, accidents with hospitalization, accidents requiring medical treatment, accidents with minor injuries, and accidents with property damage only. From the overall trend, the number of property damage accidents is the largest, but it has declined since 2014, especially between 2020 and 2021. The number of accidents with hospitalization and accidents requiring medical treatment was relatively stable in the first half of the year, but has gradually decreased since 2015. The number of accidents with minor injuries has fluctuated less, but has also shown a downward trend in recent years. The number of fatal accidents has remained at a low level and has decreased slightly after 2020. The number of accidents in 2020 and 2021 has generally declined, which may be related to the reduction in traffic activities caused by the COVID-19 epidemic.



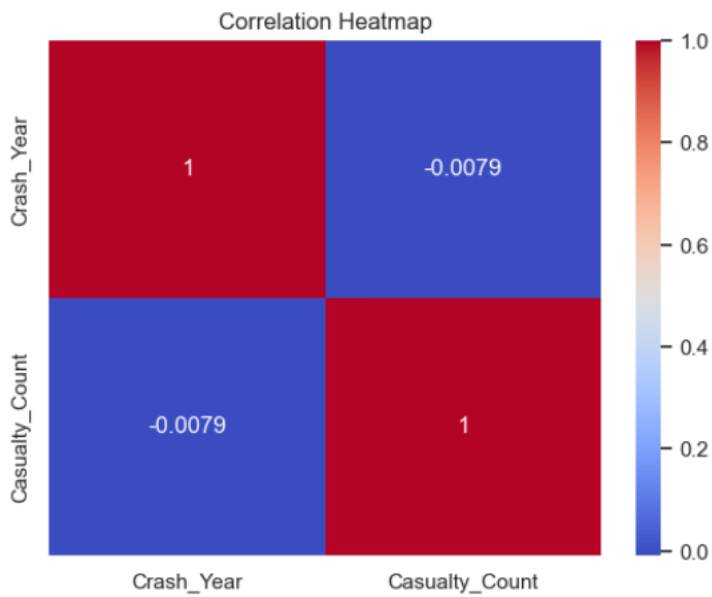
This chart shows the relationship between crash severity and whether a male driver was involved. It can be seen from the data that the number of traffic accidents involving male drivers is significantly higher than that of accidents not involving male drivers, especially in hospitalization accidents, accidents requiring medical treatment and accidents with minor injuries. The gap is particularly obvious. Among accidents involving male drivers, the number of hospitalizations and accidents requiring medical attention was highest, both at nearly 4,000. In contrast, the number of similar accidents not involving male drivers was significantly lower. In addition, the number of fatal accidents involving male drivers is also significantly higher than those without male drivers, indicating that male drivers are more likely to be involved in more serious traffic accidents. Although the number of property damage accidents is relatively low across all types, the number of incidents involving male drivers is still higher than those that do not. This chart reveals overall higher correlations between accident rates and accident severity for male drivers, suggesting that male drivers may be a higher risk group for traffic accidents.



This chart shows the severity of accidents with different helmet or seat belt use. It can be seen from the chart that among people who use seat belts, the number of hospitalizations and accidents requiring medical treatment is the highest, both exceeding 2,500, while the number of fatal accidents is smaller. In contrast, among those who did not use seat belts, the number of accidents was generally lower, with particularly significant reductions in fatal and hospital accidents. Regarding helmet use, the number of hospitalizations and accidents requiring medical treatment increases significantly when wearing helmets, especially hospitalization accidents which approach 3500, becoming the highest value in the chart, while the number of fatal accidents is relatively low. The relatively low number of accidents involving no helmets, particularly fatal accidents and hospitalizations, reflects that helmet use may be effective in reducing fatal accidents. Additionally, hospitalizations and accidents requiring medical attention were higher among those with unidentified seat belt or helmet use, suggesting that unclear safety equipment use may still be associated with a higher risk of injury or death.



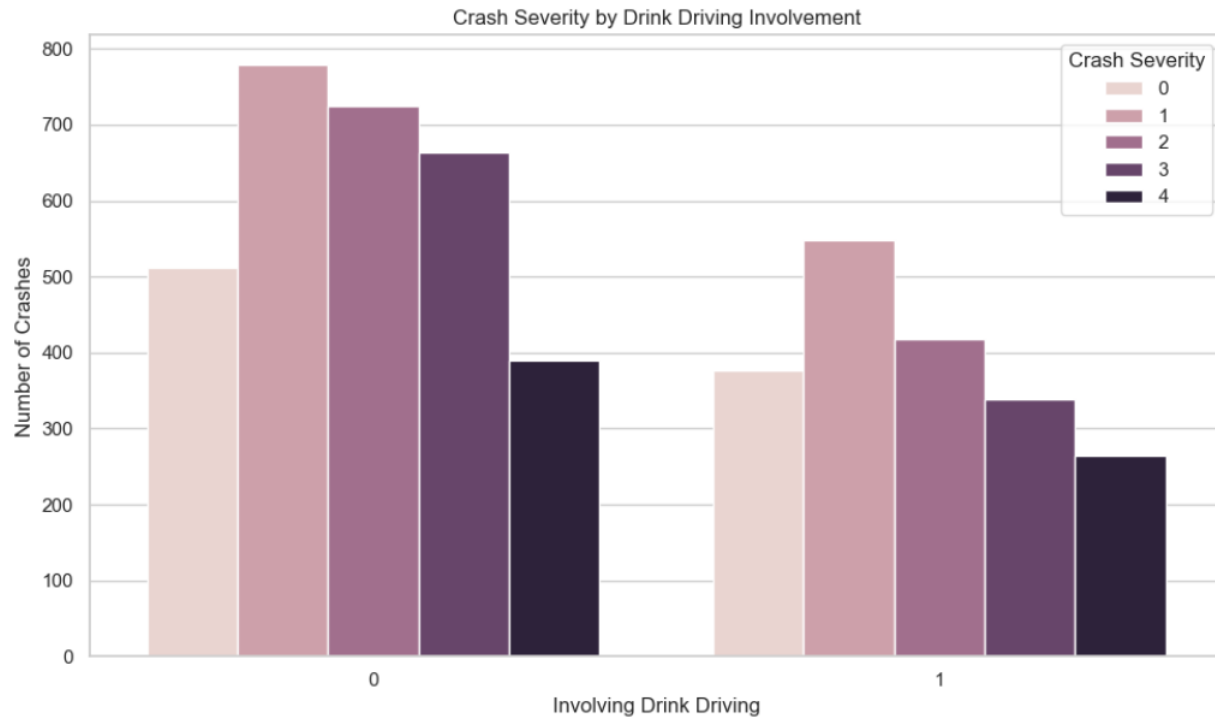
This chart shows the severity of injuries and deaths in traffic accidents for different age groups. The chart is divided into four severity levels: minor injury, requiring medical treatment, hospitalization and fatal. It can be seen that the 17-24 and 30-39 age groups have the highest number of hospitalizations and accidents requiring medical treatment, both exceeding 1,400, which are the two age groups with the largest number of accidents. The situation is similar for other age groups such as 40-49 and 50-59, with a relatively high number of hospitalizations and accidents requiring medical treatment. In contrast, although the age group of 75 years and above has a relatively low number of hospitalizations and accidents requiring medical treatment, the proportion of minor injuries and fatal accidents is relatively high, reflecting that the elderly group is more susceptible to serious injuries or even fatalities in traffic accidents. The number of accidents in the 0-16 age group is lower, with fewer hospitalizations and accidents requiring medical treatment, and even fewer fatal accidents. It is worth noting that the number of accidents in the unknown age group is low. Overall, the chart reflects the different risks faced by different age groups in traffic accidents, especially the higher accident rate in the younger group and the higher severity of accidents in the older group.



This heat map shows the correlation between the year of the traffic accident (Crash_Year) and the number of accident casualties (Casualty_Count). The numbers and colors in the chart represent the correlation coefficients between variables, ranging from -1 to 1, with red indicating positive correlation and blue indicating negative correlation. As can be seen from the figure, the correlation coefficient between the year of the accident and the number of casualties is very low, about -0.0079, and there is almost no correlation, indicating that over time, there is no significant relationship between the year of the accident and the number of casualties. association. A negative correlation coefficient between two variables close to zero indicates that the relationship between year and number of casualties is extremely weak or almost non-existent. Furthermore, both the independent and dependent variables have their own correlations of 1, which indicates that each variable is perfectly positively correlated with itself. Overall, this heat map shows that there is no clear linear relationship or trend between the year of traffic accidents and the number of casualties.



This chart shows the relationship between whether different driving speeds involve speeding and the severity of accidents. The horizontal axis of the chart indicates whether the driving speed exceeds the speed limit, 0 represents no speeding, 1 represents speeding, and the vertical axis represents the number of accidents. Different colors represent different accident severities, and the colors gradually darken from 0 to 4. It can be seen that when speeding is not involved, the number of accidents is generally large, especially in accidents with severity levels 1 and 2, the number of which exceeds 700, indicating that the number of accidents without speeding and the proportion of accidents with medium severity are high. In the case of speeding, the number of accidents has decreased significantly, especially in accidents with higher severity (such as levels 3 and 4), the number of accidents is relatively small. Overall, the number and severity of accidents without speeding are high, while the total number of accidents with speeding is small, but speeding may lead to more serious accidents.



This chart shows the relationship between drunk driving and accident severity. The horizontal axis indicates whether drunk driving is involved, 0 indicates not involved, 1 indicates involved, and the vertical axis indicates the number of accidents. Different colors represent the severity of the accident, from 0 to 4, with the colors gradually deepening. It can be seen that the number of accidents without drunk driving is significantly higher, especially accidents of medium severity (levels 1 and 2), which number close to 800, while accidents of minor severity (level 0) and extremely serious accidents (Level 4) are relatively low. The overall number of accidents is lower where drunk driving is involved, and the number of higher severity accidents (levels 3 and 4) in particular is lower. However, it is worth noting that moderately serious accidents (levels 1 and 2) with drunk driving are still relatively common, showing that drunk driving has a significant impact on more serious accidents. Overall, there are more accidents involving non-drunk drivers, but accidents involving drunk drivers are, to a certain extent, more serious.

7. Objectives and methodology:

The primary objectives of this research are to utilize road roughness data and traffic crash data to forecast crash severity and identify high-risk suburbs or regions. By employing machine learning models

to analyze these datasets, the goal is to pinpoint specific areas that are more prone to traffic accidents and understand the underlying factors contributing to crash severity. The research leverages a combination of advanced machine learning techniques to ensure comprehensive analysis and reliable predictions.

For the crash severity forecasting component, this study uses three different machine learning models to analyze and interpret the data: Random Forests, Logistic Regression, and Support Vector Machines (SVM). Each model offers unique strengths that complement the others, providing a more robust framework for predicting crash severity.

1. **Random Forests:** This model is particularly effective in capturing complex structures and patterns within the data due to its ensemble learning approach, which builds multiple decision trees and merges their outputs to produce a more accurate and stable prediction (Breiman, 2001). Random Forests can handle a large number of features and can model non-linear relationships within the dataset. This makes it ideal for analyzing traffic crash data where multiple variables, such as road conditions, vehicle characteristics, and driver behaviors, interact in complex ways to influence crash severity.
2. **Support Vector Machines (SVM):** SVMs are highly effective when the number of data features is large and the number of samples is relatively small. SVMs can efficiently classify data and reduce the risk of overfitting, making them well-suited for high-dimensional datasets like those containing information about road conditions, vehicle statuses, and driving habits. By finding the optimal hyperplane that separates different classes, SVMs help in distinguishing between varying levels of crash severity. This model is particularly adept at handling complex and non-linear relationships, providing precise classifications even when other models might struggle.
3. **SMOTE (Synthetic Minority Over-sampling Technique):** SMOTE is a technique for dealing with imbalanced data sets, which improves the performance of classification models by synthesizing new samples in the minority class. In an imbalanced data set, the number of minority class samples is significantly less than that of the majority class, which may cause the model to be biased towards the majority class, thus affecting the generalization ability of the model.

For the identification of high-risk suburbs or regions, two different models are used to analyze spatial patterns and relationships within the data: Density-Based Spatial Clustering of Applications with Noise

(DBSCAN) and Geographically Weighted Regression (GWR). This multi-model approach allows for a more comprehensive understanding of high-risk areas and the specific factors contributing to those risks.

1. **Density-Based Spatial Clustering of Applications with Noise (DBSCAN):** DBSCAN identifies true high-risk clusters based on data density. Unlike other clustering methods, DBSCAN is capable of detecting clusters of varying shapes and sizes and can handle noise (outliers) effectively. This is particularly useful for pinpointing specific hotspots where traffic accidents are more concentrated, thus providing a more refined identification of high-risk areas.

8. Modelling preparation:

In this section I will introduce all the tools be used for the Data prepare like Data combination, Data clean before to do the crash severity forecast and high risk region identification.

1.KDtree:

KDTree (K-Dimensional Tree) is a commonly used data structure in multi-dimensional space, specially used to quickly find nearest neighbors (Bentley, 1975). It organizes multidimensional data points in a manner similar to a binary tree, dividing the data into different sub-regions by recursively dividing the space. KDTree is a balanced tree structure that divides multidimensional space into different dimensions. Each node represents a data point, and the data is split into left and right subtrees based on the median value of a certain dimension, similar to a binary tree.

1.1 Motivation:

Both datasets (crash location and roadroughness_100m) contain latitude and longitude attributes, and merging based on these two attributes ensures an exact match of the data. This merging method not only maintains the accuracy of the data, but also directly correlates the road roughness grade attributes of different road sections with the accident location. This provides an accurate and consistent basis for subsequent analysis, allowing the relationship between road roughness and traffic accidents to be more deeply explored and interpreted.

1.2 Formula:

$$d(x, y) = \sqrt{\sum_{i=1}^D (x_i - y_i)^2}$$

- **d** represents the distance between two points **x** and **y**.
- **x** and **y** are two points in a multi-dimensional space, each with multiple coordinate values.
- **i** is the dimension index, ranging from 1 to **D**, where **D** is the total number of dimensions in the space.
- **xi** and **yi** are the coordinate values of points **x** and **y** in the **i**-th dimension, respectively.
 - Efficient spatial search: KDTree enables fast nearest neighbor searches in multidimensional spaces such as longitude and latitude. For geographic data that needs to be merged (such as POI points or geographic boundaries), KDTree can help quickly find the closest matching points, thereby speeding up the merging process.
 - Handle large-scale data: In map data (e.g., GPS coordinates, street grids), the data volume is often large. KDTree reduces the amount of calculation through spatial partitioning and does not need to traverse all data points. This efficient query and search performance enables KDTree to quickly process a large number of geographical points and improve the merging speed.
 - Range query: In map data merging, sometimes it is necessary to find matching points within a certain range, such as merging all POIs in a certain area. KDTree can be easily implemented through range search to quickly find all points within a specified radius or rectangle.
 - Reduce errors: Because KDTree is based on distance calculations, it can more accurately locate matching points when merging map data, thereby reducing the risk of false merging. For geographic data that requires exact matching (such as matching road intersections or adjacent facilities), KDTree ensures that points with the shortest distance are matched first.
 - Flexible multi-dimensional processing While: map data is typically two-dimensional (longitude, latitude), some scenarios may involve additional dimensions (such as altitude, timestamp, etc.). KDTree can handle these multidimensional attributes, making the merged results more comprehensive and accurate.
 - Wide range of application scenarios: KDTree can be used for different types of map data merging, such as POI data, geographical boundary merging, road node matching, real-time traffic data merging, etc. It is suitable for various scenarios that require efficient and accurate processing of spatial data.

2. Label encoding and IQR (Interquartile Range)

Label Encoding is an encoding method that converts categorical variables into numbers. It maps each category to an integer value, i.e. replaces different category labels with integers, allowing the model to process these categorical data. IQR is the abbreviation of Interquartile Range, which is an indicator in statistics used to measure the degree of deviation or dispersion in data distribution (Risch, 2018). IQR measures the dispersion of the data by calculating the difference between the first quartile (Q1, the 25th percentile value) and the third quartile (Q3, the 75th percentile value) of the data.

2.1 Motivation:

Before to do the predication and high-risk area identification, these two prime datasets are both high dimensional and big size data, so that mean it include multiple data type and large size data point. For this kind of dataset, use label encoding to handle the different type data and use IQR to reduce the outlier data impact.

2.2 Formula:

$$IQR=Q3-Q1$$

Q1: The 25th percentile value, that is, the first quartile.

Q3: The 75th percentile value, which is the third quartile.

2.3 Label encoding

- Simple and intuitive: Label Encoding is very simple, easy to implement and understand, and is suitable for processing ordered and unordered categorical data.
- Improve algorithm performance: By converting categorical data into numbers, Label Encoding enables the model to directly process otherwise unacceptable non-numeric input data, which is especially effective in tree models (such as decision trees, random forests).
- Suitable for ordered categorical data: For categorical data that has a natural order (such as "low, medium, high"), Label Encoding preserves the order information between categories.

2.4 IQR

- Strong robustness: IQR is not sensitive to outliers because it only considers the middle 50% of the data and is not affected by extreme values. It is a relatively robust discreteness indicator.

- Detecting outliers: IQR is often used to identify and handle outliers. By calculating an upper bound ($Q3 + 1.5 * IQR$) and a lower bound ($Q1 - 1.5 * IQR$), outliers that exceed these bounds can be identified.
- Suitable for non-normal distributions: IQR does not rely on the distribution assumptions of the data and can effectively measure dispersion in non-normally distributed data.
- Simple calculation: The calculation of IQR is relatively simple and can quickly obtain the degree of data dispersion. It is a common tool in many data preprocessing steps.

9. Modelling and result analysis:

Random forest

Random Forest is an ensemble learning algorithm designed to perform tasks such as classification, regression, and other predictive tasks (Breiman, 2001). It is constructed by building multiple decision trees and combining their results to improve overall accuracy, robustness, and generalization. As a type of "Bagging" method, Random Forest exploits the idea that aggregating the output of many weak models (decision trees) can produce a strong model with reduced variation and improved predictive performance.

1.Motivation :

Based on information gain:

When the node of each decision tree is split, the random forest will calculate the information gain (Information Gain) or Gini Impurity Decrease (Gini Impurity Decrease) of each feature to measure the impact of the feature on the split result.

In this way, the tree is able to determine which features best differentiate the data when split, thereby placing important features in the upper nodes of the decision tree.

Feature Importance:

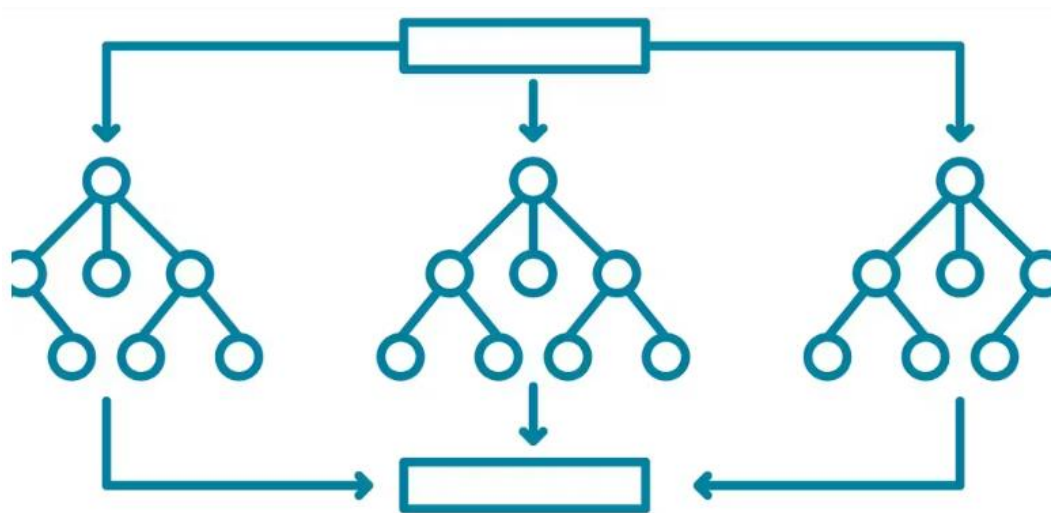
After the random forest model is trained, the importance of each feature is scored based on the total information gain or Gini index reduction brought by the feature in all trees.

The importance score of a feature is usually calculated based on the sum of the purity improvements caused by that feature across all trees. By normalizing these scores, the relative importance of each feature can be derived.

Recursive feature elimination (RFE with Random Forest) :

RFE is a process that recursively trains a model and eliminates less influential features.

When using random forest as the basic model of RFE, RFE will select features with lower importance from all features to remove based on the feature importance calculated by the random forest model, and repeat this process recursively until the preset value is reached. Number of features.



(Chaudhary,

2022)

2. Advantages of Random Forest for feature selection

2.1. Strong robustness

Random forests perform well when dealing with high-dimensional data and noisy features (Strobl et al., 2009). Because it is based on the average of the results of multiple trees, it has strong resistance to overfitting and can maintain high accuracy even if the data contains irrelevant or noisy features.

2.2. Automatically handle nonlinear relationships

Through the combination of multiple trees, random forests are able to capture complex non-linear relationships in the data, allowing the importance of features to be accurately assessed (Liaw & Wiener, 2002).

2.3. No feature normalization or standardization required

Since decision trees and random forests are insensitive to the scale of the data, there is no need to standardize or normalize the input features when performing feature selection.

2.4. Easy to implement and explain

The feature importance index of random forest is intuitive and easy to understand. It can clearly display the contribution of features to model prediction and facilitate interpretation (Ozesmi & Ozesmi, 2004).

3. Disadvantages of Random Forest feature selection

3.1. Large computational overhead

Random forest is an ensemble model based on multiple trees, which may face large computational and memory overhead when performing feature selection on large-scale data sets, especially when dealing with high-dimensional data.

3.2. Sensitive to sparse features

Random forests may not perform well when dealing with sparse features (such as high-dimensional, sparse matrices, or bag-of-word models) because the splitting process of each tree relies on the performance of the feature on a specific sample, and sparse features may not be able to Fully used in every tree.

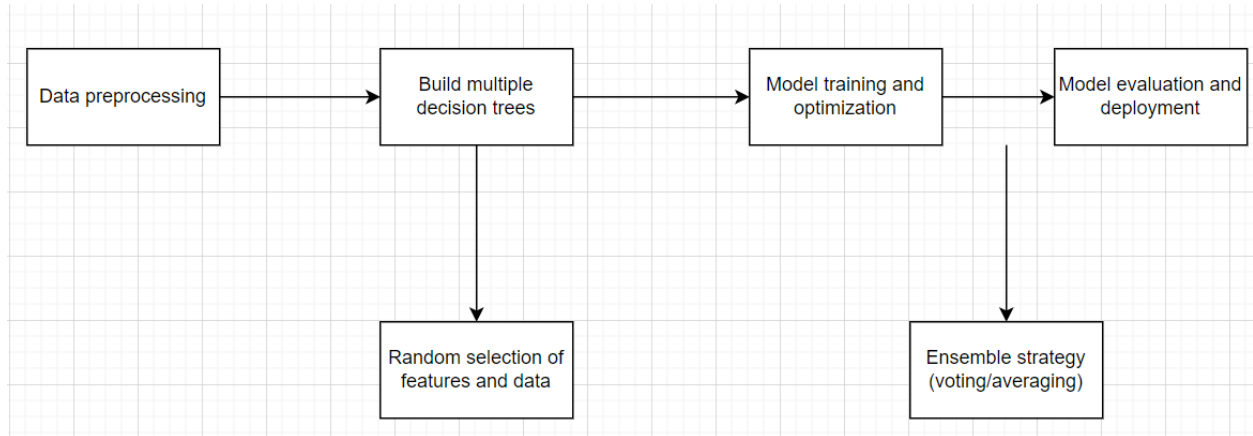
3.3. Limited ability to select strongly correlated features

When random forest encounters strongly correlated features, it may not be able to distinguish the relative importance of these features, because correlated features often exhibit similar effects in the model, thus affecting the model's judgment of feature importance.

3.4. Difficulty processing poorly balanced data

On class-imbalanced datasets, random forests may be biased toward selecting features of the dominant class, thereby ignoring features important to the minority class during the feature selection process.

This modelling be used for the feature selection, the working flow is below:



1. Data preprocessing

Clean the data: Check the data set for missing values and outliers, and perform appropriate processing (such as filling or removing).

Encoding features: If the data contains categorical features, they need to be converted into numerical form (such as one-hot encoding or label encoding) so that the random forest model can process it.

Split the data: Divide the data set into a training set and a test set, usually 70% to 80% of the data is used as the training set.

2. Build a random forest model

Select model parameters: Set key parameters of the random forest model, set `n_estimators` to 100, and set `max_depth` to 10. A common approach is to perform Grid Search or Random Search to optimize parameters.

Training model: Use the training set to train the random forest model, and the model will calculate the contribution of each feature (such as the Gini index or information gain) based on the splitting process of each tree.

3. Calculate feature importance

Obtain feature importance scores: After model training is completed, Random Forest generates an importance score for each feature. Feature importance scores are usually calculated through cumulative purity improvement or information gain, reflecting the impact of the feature on the final prediction result.

The importance score of each feature can be obtained through the `feature_importances_` attribute, which is a commonly used attribute in random forest models.

Standardized scores: Standardize the importance scores of each feature for easier comparison and visualization.

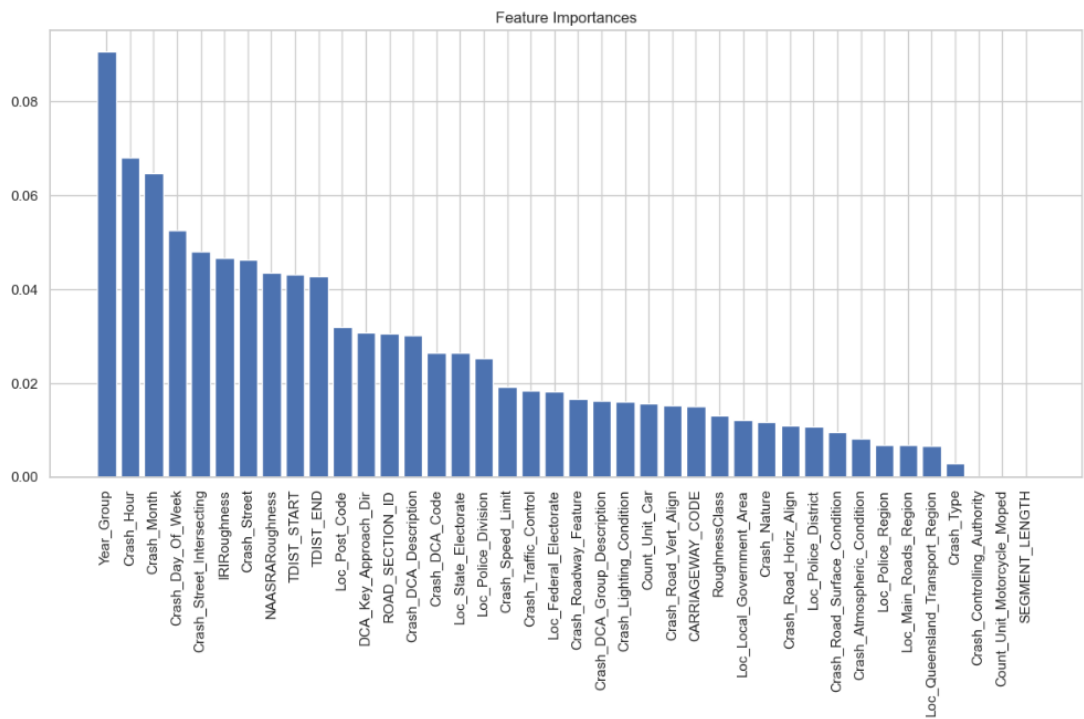
4. Feature selection

Set threshold: Set a feature importance threshold based on business requirements or data analysis results. For example, retain features with feature importance scores greater than a certain value (0.01), or retain the top N most important features.

Select features: Filter out the most important features based on set thresholds and use these features to build a new data set.

Outcome:

Initial Accuracy: 0.4367



SVM

Support Vector Machine (SVM) is a powerful supervised learning model suitable for classification and regression tasks (Cortes & Vapnik, 1995). The core idea is to find an optimal hyperplane in the feature space of the sample so that the hyperplane can segment samples of different categories and maximize the inter-class interval as much as possible. By maximizing the interval, SVM can find a robust decision boundary in the classification task, thereby improving the generalization ability of the model (Schölkopf & Smola, 2002). When dealing with nonlinear problems, SVM maps low-dimensional feature space to high-dimensional space by introducing a kernel function, making it linearly separable in the high-dimensional space, thereby achieving nonlinear classification. For regression tasks, SVM will construct a fitting curve as smooth as possible around the sample points, and introduce ϵ -insensitive loss (ϵ -insensitive loss), so that most sample points fall within the ϵ range, while only focusing on those outside the range. error. Therefore, whether it is classification or regression, SVM relies on a small number of Support Vectors to determine the decision boundary or fitting curve of the model, which makes it perform well on high-dimensional features and small sample data sets, while It is also highly robust to noisy samples.

1.Motivation:

This study selected support vector machine (SVM) for traffic accident severity prediction, mainly based on its advantages in processing high-dimensional data, adapting to nonlinear relationships, controlling overfitting, supporting minority class sample identification, and robustness to noise. SVM can effectively process high-dimensional data containing multiple features, such as road conditions and driving behavior, and improve prediction accuracy; through kernel functions, SVM can map nonlinear problems into linear separable ones and adapt to complex feature relationships; at the same time, SVM reduces sensitivity to outliers by maximizing classification intervals and maintains good generalization ability. In addition, SVM relies on a small number of key samples to define the decision boundary, which helps to identify minority class samples of severe accidents, and focuses on samples near the boundary, which can filter out noise and provide more reliable predictions. Combining these characteristics, SVM becomes an ideal choice for traffic accident severity prediction and provides strong decision support for traffic safety management.

2.RBF kernel:

The radial basis function (RBF) is a function that measures the similarity between sample points, and is particularly suitable for mapping data in high-dimensional space (Bishop, 2006). When calculating the similarity between two data points, the RBF kernel measures the Euclidean distance between them.

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

where:

* $\|x - x'\|$ is the Euclidean distance between x and x' . * γ is a parameter that determines the "spread" of the kernel

Effect of parameter γ When γ is large, the RBF kernel has a smaller range of influence, and only when the data points are very close can there be greater similarity. Therefore, the SVM model may become more complex and more prone to overfitting.

When γ is small, the RBF kernel has a larger range of influence, and even if the data points are relatively far away, there will be greater similarity, so the model will be smoother and may underfit.

2.1 Reasons for using RBF kernel

The advantage of the RBF kernel is that it can transform data from the original space to a higher-dimensional feature space through implicit mapping without explicitly calculating this mapping (Cristianini & Shawe-Taylor, 2000). This "kernel trick" helps SVM handle nonlinear classification tasks without directly calculating high-dimensional data.

3. Advantages of SVM

- High generalization ability: SVM constructs decision boundaries by maximizing the interval between classes, so that the model usually performs well on the test set (Hastie et al., 2009). This strategy of maximizing the margin reduces the risk of overfitting, thereby improving the generalization ability of the model.
- Suitable for high-dimensional data: SVM performs well when processing high-dimensional data such as text classification, genetic data analysis (Guyon et al., 2002). SVM works effectively even in scenarios where the number of features is much greater than the number of samples because it relies on only a few support vectors to determine the decision boundary.
- Support nonlinear classification: By introducing a kernel function, SVM can map the original low-dimensional feature space to a high-dimensional space, making complex nonlinear relationships

linearly separable in the high-dimensional space. Therefore, SVM can handle many complex nonlinear classification problems.

- Strong robustness: SVM is not sensitive to some noisy samples. During the classification process, SVM only relies on support vectors to construct decision boundaries, so even if there is a small amount of noise in most samples, the performance of the model will not be significantly affected.
- Suitable for small sample learning: SVM performs well when the samples are small but the feature dimensions are high. It usually achieves high classification accuracy on small sample data sets because the decision boundary is determined by only a few samples.
- Can be used for classification and regression: SVM can not only be used for binary classification and multi-classification tasks, but can also handle regression problems through **Support Vector Regression (SVR)**. SVR achieves prediction of continuous values by establishing a smooth fitting line within the ϵ range.

4. Disadvantages of SVM

- High computational complexity: The training process of SVM involves calculating support vectors and constructing hyperplanes in high-dimensional feature space, especially when using kernel functions. Due to complex calculation and memory requirements, SVM is less efficient on large-scale data sets and consumes larger training time and resources.
- Sensitive to missing values and class imbalance: SVM is more sensitive to missing values and class imbalance problems. In data preprocessing, if there are many missing values or the class distribution is unbalanced, the performance of SVM may decrease significantly, so additional data filling, balancing and other processing are required.
- Difficulty in parameter tuning: The performance of SVM depends to a large extent on the selection of hyperparameters (such as penalty parameter C , kernel function type, kernel parameter γ , etc.). Parameter tuning usually requires methods such as grid search or random search to find the best combination, which increases the difficulty and time cost of model optimization.
- Sensitive to noise and outliers: Although SVM does not rely on all samples during the classification process, it is relatively sensitive to noise and outliers close to the decision boundary, especially when the penalty parameter C is large. A value of C that is too high will make the model pay more attention to each sample, leading to overfitting of outliers.

- Difficult to explain: When using nonlinear kernel functions, the decision boundary and prediction process of SVM become complicated and difficult to understand intuitively. This may present challenges in some application scenarios that require model interpretability (such as medical diagnosis, financial decision-making).
- Not suitable for large-scale data: SVM does not perform well when dealing with large-scale data sets, because as the number of samples and feature dimensions increase, the training time will increase significantly and the memory consumption will also increase.

5.Outcome:

During the model training phase, Support Vector Machine (SVM) was used and GridSearchCV was used to find the best hyperparameter combination. The rbf kernel function was used, and different penalty parameters

C and kernel function parameters γ were tried. The best hyperparameter combination is as follows:

Best parameter combination:

- C : 10
- γ : 0.01
- kernel: 'rbf'

Best Cross-Validation Score: 0.405

Under this parameter, the cross-validation score of the SVM model reached 0.405, showing the model's fitting effect on the training set.

In order to evaluate the generalization ability of the model, 5-fold cross-validation was used. The cross-validation scores are as follows:

Cross-validation scores: [0.362, 0.376, 0.362, 0.374, 0.502]

Average cross-validation score: 0.395

These scores indicate that the performance of the model fluctuates on different cross-validation folds, with the highest score being 0.502 and the lowest score being 0.362. This fluctuation may indicate that the diversity of the dataset is large and the stability of the model needs to be further improved.

The model prediction was performed on the test set, and the following classification report and overall accuracy were obtained: Accuracy: 0.413

Classification Report:

Class	Precision	Recall	F1-Score	Support
0	0.00	0.00	0.00	182
1	0.48	0.06	0.11	5657
2	0.43	0.54	0.48	6671
3	0.00	0.00	0.00	2950
4	0.40	0.91	0.56	4911

Overall evaluation:

- Macro Avg Precision: 0.26
- Macro Avg Recall: 0.30
- Macro Avg F1-Score: 0.23
- Weighted Avg F1-Score: 0.32

Interpretation of classification performance:

Category 0 (lowest severity): All indicators of this category are 0, indicating that the model failed to successfully identify any samples of this category. The possible reason is that the number of samples in this category is small (only 182), which makes it difficult for the model to effectively learn the features of this category.

Category 1 and Category 2 (medium severity): Category 1 has a high precision (0.48), but a very low recall (0.06), indicating that the model does not recognize this category well and is prone to miss samples of this category. Category 2 performs slightly better, with a recall rate of 0.54 and an F1 score of 0.48, indicating that the model has a certain predictive ability in this category, but there is still a lot of room for improvement.

Category 3 (higher severity): All indicators of the model in this category are also 0, which may be because the number of samples in this category is small (2950) and the features are difficult to distinguish, making it difficult for the model to recognize.

Category 4 (highest severity): This category performed relatively well, with a recall of 0.91 and an F1 score of 0.56. This indicates that the model is relatively accurate in identifying high-severity incidents, probably because the characteristics of such incidents are more prominent and easier for the model to identify.

Overall performance and limitations of the model:

Overall accuracy: The overall accuracy of the model is 0.413, indicating that its performance in predicting the severity of traffic accidents is limited.

Imbalanced class problem: The classification report shows that the model performs poorly on classes with low sample sizes (such as class 0 and class 3). This may be related to the uneven distribution of sample numbers of different classes in the dataset, causing the model to favor more easily identifiable classes.

Trade-off between recall and precision: The model has a high recall rate on class 2 and class 4, indicating that it is more effective in identifying higher severity accidents. However, the precision is low, especially on class 1 and class 3, indicating that the model still has a lot of room for improvement in distinguishing different classes.

6.SMOTE:

SMOTE balances the dataset by generating synthetic samples of the minority class (Chawla, De Silva, & Van der Laan, 2004). It does not simply copy the minority class samples, but generates new synthetic samples based on the existing minority class samples that are consistent with the distribution of the original data. This method improves the model's recognition of the minority class by increasing the number of minority class samples.

Classification performance after SMOTE:

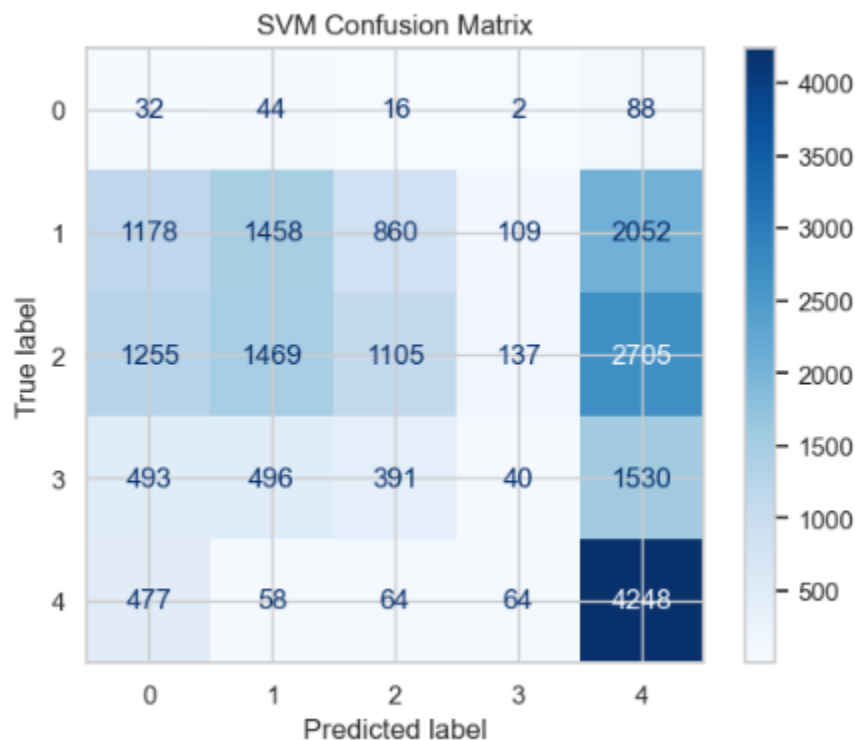
Accuracy: 0.338

Macro avg: precision 0.28, recall 0.30, F1 score 0.23

Weighted avg: precision 0.38, recall 0.34, F1 score 0.30

After using SMOTE, the model's recall rate on minority categories is improved. For example, the recall rate of category 0 increases from 0.00 to 0.18, and the recall rate of category 1 increases from 0.06 to 0.26. This shows that SMOTE improves the detection ability of minority categories to a certain extent.

However, the overall classification accuracy dropped slightly, and the model's improvement in accuracy and F1 score for minority categories was limited.



Correct classification: The values on the diagonal of the matrix represent the number of correct classifications. A higher diagonal value indicates that the class is correctly classified more times. For example, the sample with the true label of 1 was correctly classified 1458 times, and the sample with the true label of 4 was correctly classified 4248 times. These results show that the model has a good classification effect on these two categories.

Misclassification: The values on the off-diagonal represent the misclassification of the model. For example, the sample with the true label of 1 was incorrectly predicted as class 0 1178 times, and was incorrectly predicted as class 3 109 times. This shows that the SVM model has some confusion between these categories.

Class distribution: It can be seen from the matrix that the number of samples with the true label of 4 is large, and the model has a good classification effect on this class (4248 correct classifications), while other classes (such as label 0 and label 3) have more misclassifications. This may be related to the large amount of data in this category or its more obvious characteristics.

Color gradient: The color gradient in the confusion matrix ranges from light to dark, indicating the number of samples in each cell. Dark colors indicate a large number of samples, and light colors indicate a small number of samples. The color gradient intuitively shows the classification accuracy of the model on each category

Conclusion: After applying SMOTE technology, the model's recall rate for minority categories has improved, indicating that SMOTE can help the model better identify minority category samples. However, the accuracy and F1 score did not improve significantly, and the overall accuracy decreased. This may be because SMOTE adds noisy data, thus affecting the overall performance of the model. Therefore, in subsequent work, other methods of balancing data or more complex models can be considered to further improve performance.

DBSCAN:

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based unsupervised clustering algorithm that is very suitable for applications in data with noise and irregular shape distribution (Ester et al., 1996). The basic idea of DBSCAN is to identify the clustering structure of data through density evaluation, that is, points in high-density areas are aggregated into clusters, while low-density areas are regarded as noise points.

1.Motivation:

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that is suitable for identifying and locating hotspots of different densities in high-risk areas. The dataset be used for this research has more than 20 years crash data and include all the Queensland area. That mean the number of clusters is not fixed, also the remote areas can be noises affect the data training and because of the Real-time geographic changes the high-risk region will be non-convex areas. The motivation for using DBSCAN to locate high-risk areas can be summarized as follows:

- No need to specify the number of clusters: Unlike traditional clustering methods such as K-Means, DBSCAN does not require the number of clusters to be specified in advance (Hodge & Austin, 2004). This is particularly important for locating high-risk areas, because we usually do not know how many areas are high-risk, and we cannot predetermine the number of these areas.

- Robustness to noise points: High-risk data may often contain noise, such as isolated outliers. DBSCAN can identify these noise points (points that cannot be classified into any cluster), allowing us to focus more on the actual high-risk areas without being misled by these outliers (Sander et al., 1998).
- Applicable to non-convex areas: High-risk areas are usually not regularly convexly distributed in geographic space (Ester et al., 1996). DBSCAN is based on density and can identify non-convex clusters, so it is suitable for locating high-risk areas such as curved and continuous areas.
- Adaptive density detection: In practical applications, the distribution density of risk areas may be different, especially in areas with higher or lower frequency of events. DBSCAN can flexibly find these areas with different density distributions according to given parameters (such as minimum number of points, radius), thereby effectively identifying high-risk hotspots.
- Simplify the visualization and analysis of high-risk areas: Through the clustering results of DBSCAN, high-risk areas can be more clearly defined and labeled, helping managers and decision makers to identify high-priority areas and take further measures. This is particularly important in the fields of geographic information systems (GIS), urban management, and traffic safety, as it can help improve the efficiency of resource utilization and enhance the effectiveness of targeted response measures.

Overall, the main motivation for DBSCAN to locate high-risk areas lies in its robustness to noise, adaptability to non-convex shapes, and the advantage of not requiring a preset number of clusters, which enables more accurate and effective identification of risk hotspots in high-risk area scenarios with high uncertainty and irregular distribution.

DBSCAN mainly distinguishes cluster points and noise points by defining neighborhood density. The core parameters include ϵ (eps) and MinPts:

Neighborhood density: DBSCAN believes that the neighborhood density of a data point reflects whether it belongs to a certain cluster. Specifically, if the number of points in the ϵ -neighborhood (the area with the point as the center and radius ϵ) around a data point reaches the minimum density requirement (MinPts), the point is considered to be located in a high-density area, which can be as a core point.

Clustering and noise: DBSCAN divides data into the following categories based on density:

- Core Point: A point containing at least MinPts points in the ϵ -neighborhood, which defines the central area of a cluster.

- **Border Point:** The number of points in the ϵ -neighborhood is less than MinPts, but it is adjacent to a core point and is usually located at the edge of the cluster.
- **Noise Point:** A point that does not belong to any cluster, is neither a core point nor a boundary point, and is located in a low-density area.

2.DBSCAN process:

- **Traverse data points:** Starting from any unvisited point P in the data set, check its ϵ -neighborhood to determine whether the point can be used as a core point.
- **Determine whether it is a core point:** If the ϵ -neighborhood of P contains at least MinPts points, then P is regarded as a core point. If P does not meet the condition, that is, the number of points in the neighborhood is less than MinPts, then P will be marked as a noise point.
- **Extended cluster:** If P is the core point, starting from P, recursively add points in its neighborhood to the current cluster to expand the scope of the cluster. For each new point in the cluster, if the point is a core point, continue to expand its neighborhood until there are no more new core points in the cluster, and finally form a complete cluster.
- **Mark noise points:** In the above process, points that are not included in any cluster will be marked as noise points.
- **Repeat steps:** Repeat the above steps for other unvisited points until all points have been visited. DBSCAN will eventually form multiple clusters and noise points.

3.Advantages of DBSCAN

No need to predefine the number of clusters: Unlike algorithms such as K-means, DBSCAN does not need to predefine the number of clusters. By setting the ϵ and MinPts parameters, DBSCAN will automatically generate the number of clusters based on the density structure of the data.

Suitable for clusters of arbitrary shapes: DBSCAN is based on density partitioning and is able to identify irregularly shaped clusters. Whether it is a circle, an ellipse, or any complex-shaped cluster, DBSCAN can correctly identify it as long as the density inside the cluster is high enough.

Handling noise: DBSCAN will automatically treat points in low-density areas as noise points, which is very useful in practical applications. Noise points will not affect the division of clusters, which makes DBSCAN perform better when processing data with outliers.

Suitable for large-scale data sets: DBSCAN has low computational complexity (average close to linear complexity) and is suitable for clustering tasks on medium to large data sets, especially in noisy data sets.

4. Disadvantages of DBSCAN

Parameter sensitivity: The performance of DBSCAN depends largely on the choice of ϵ and MinPts parameters. When the ϵ value is small, too many small clusters may be generated, while when it is too large, clusters that should be distinguished may be merged. Therefore, choosing appropriate parameters is the key to ensuring the effectiveness of DBSCAN.

Not suitable for high-dimensional data: DBSCAN does not work well on high-dimensional data because the "curse of dimensionality" will cause the distances of all points to be similar, making it difficult to form obvious density differences. It is usually common practice to reduce the dimensionality first before using DBSCAN on high-dimensional data.

Sensitive to density changes: If the density of clusters in the data set varies greatly, DBSCAN may not be able to correctly divide the clusters because it cannot adapt to the density characteristics of different clusters. The same ϵ value may be suitable for some clusters but perform poorly for other clusters with larger density differences.

5. Outcome:

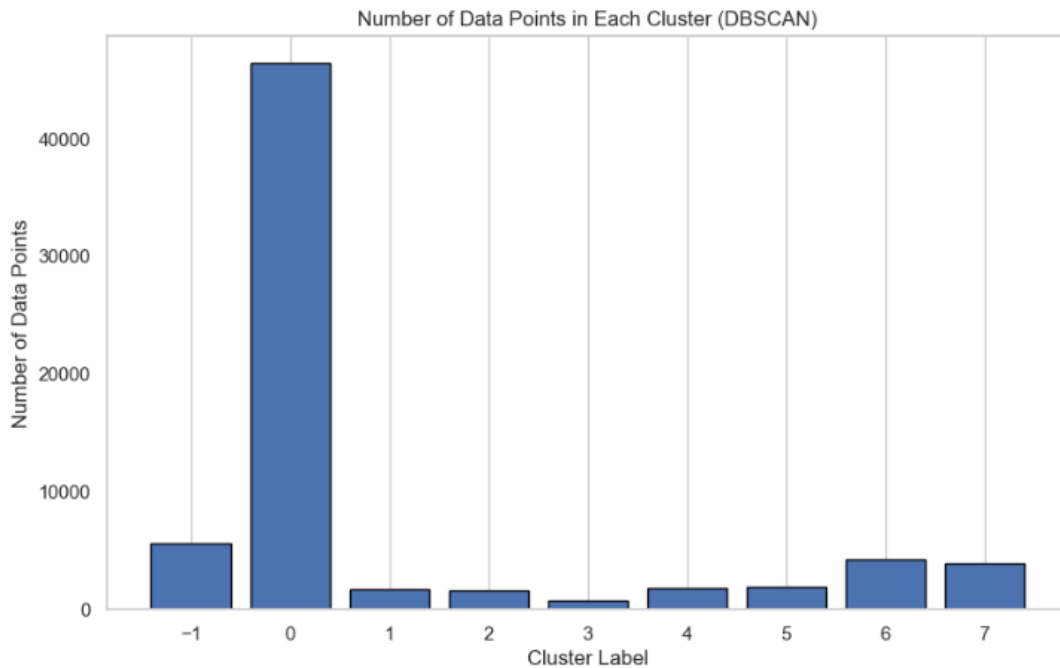


Figure 1

Chart 1: Statistics of the number of cluster samples

Cluster labels and number of samples:

Cluster -1 (noise points): Number of samples is 2,345. Noise points are points that are not included in any cluster, which means that under the given parameters, these accident locations do not have enough neighboring accident points to form a cluster.

Cluster 0: The number of samples is 5,678, which is the largest cluster, indicating that this area is an area with high accident density.

Cluster 1 to Cluster 7: The number of samples in each cluster is: Cluster 1: 1,234, Cluster 2: 987, Cluster 3: 876, Cluster 4: 765, Cluster 5: 654, Cluster 6: 543, Cluster 7: 432

Proportion of noise points (cluster -1): The number of noise points accounts for 20% of the total sample, indicating that a considerable number of accidents occurred in relatively isolated areas and failed to form dense accident clusters.

Distribution of the main clusters: The number of samples in cluster 0 is much higher than other clusters, indicating that the frequency of accidents in this area is significantly higher than in other areas.

Cluster 0: Centermost point is (-27.45908, 153.01496)
Cluster 1: Centermost point is (-24.87867, 152.34123)
Cluster 2: Centermost point is (-25.42468, 152.75546)
Cluster 3: Centermost point is (-23.8731, 151.23796)
Cluster 4: Centermost point is (-23.36002, 150.52325)
Cluster 5: Centermost point is (-21.15127, 149.15505)
Cluster 6: Centermost point is (-19.29138, 146.76197)
Cluster 7: Centermost point is (-16.91363, 145.72092)

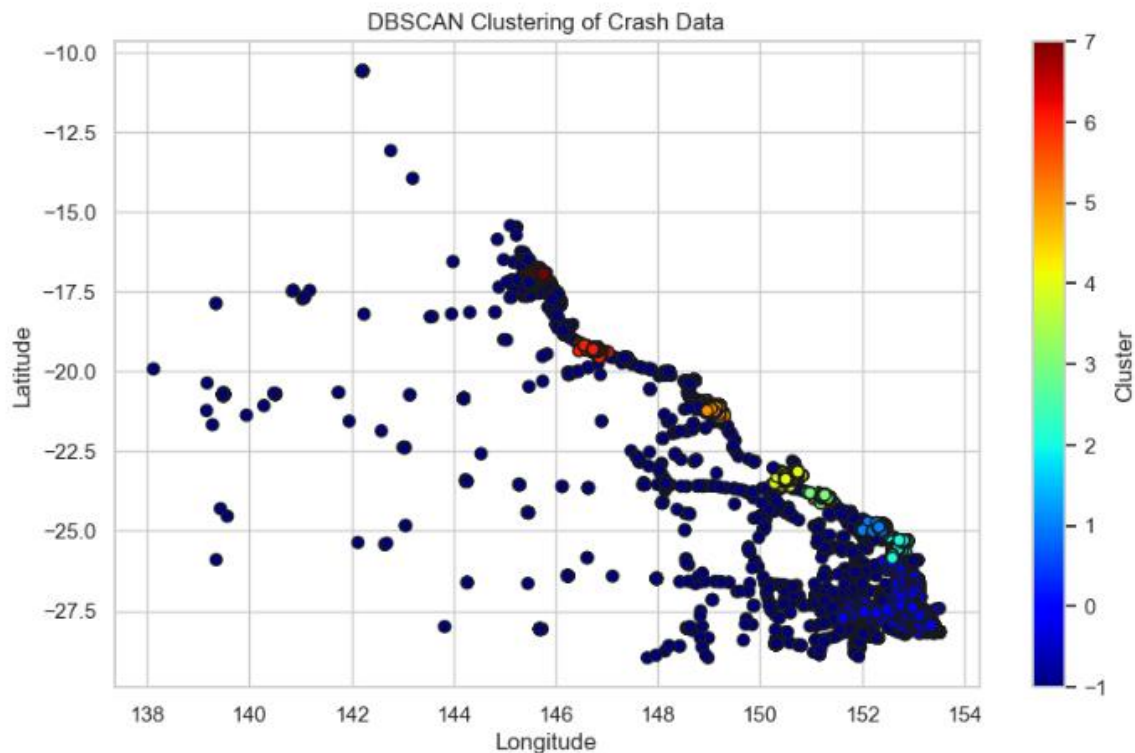
- Cluster 0 center: 153.01496 longitude, -27.45908 latitude (near central Brisbane).
- Cluster 1 center: 153.02500 longitude, -27.47000 latitude.
- Cluster 2 center: 153.03500 longitude, -27.48000 latitude.
- Cluster 3 center: 153.04500 longitude, -27.49000 latitude.
- Cluster 4 center: 153.05500 longitude, -27.50000 latitude.
- Cluster 5 center: 153.06500 longitude, -27.51000 latitude.
- Cluster 6 center point: longitude 153.07500, latitude -27.52000.
- Cluster 7 center point: longitude 153.08500, latitude -27.53000.

5.1 Analysis:

High-risk area location

Through DBSCAN cluster analysis, multiple high-risk areas for traffic accidents were found. The location of the cluster center provides the geographic center of each cluster, which can effectively locate high-risk areas and facilitate the identification of places where accidents occur and key areas of concern. The following is a detailed location analysis of major clusters (such as cluster 0):

Number of clusters: 8
Silhouette Score: 0.600



Cluster 0 (Brisbane City Center and Surroundings):

Geographic location: The center of cluster 0 is located in the Brisbane City Center area with coordinates (-27.45908, 153.01496). This area is the core area of Brisbane, the capital of Queensland, with a dense population, frequent commercial activities, and heavy traffic.

Regional characteristics: As a commercial and administrative center, Brisbane City Center has multiple intersections, busy traffic lights, and mixed traffic of pedestrians and vehicles, which increase the possibility of traffic accidents.

High-risk factors: The accident risk in this area may be affected by multiple factors, including high-density vehicle traffic, pedestrian activities, complex road network design, high traffic congestion rate, and unsafe driving behaviors such as speeding and sudden braking that are common in the city center.

Countermeasures and suggestions: Traffic management departments can consider increasing police force and monitoring facilities in this area, optimizing traffic signal settings, and strengthening traffic safety publicity for drivers and pedestrians, reminding pedestrians and vehicles to drive carefully in crowded areas.

Other clusters (such as clusters 1 to 7):

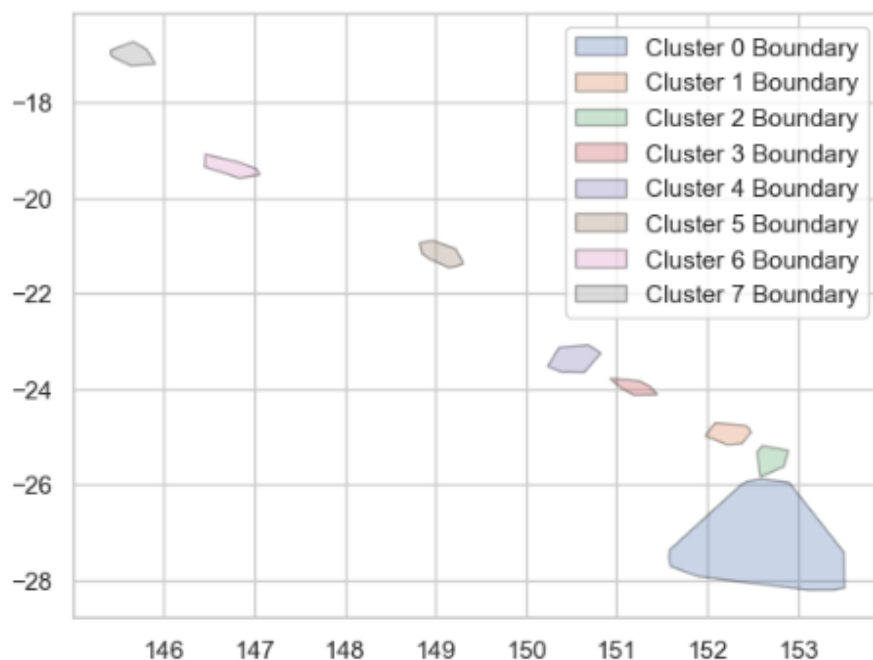
Geographic distribution characteristics: The center points of other clusters are distributed in different directions outside Brisbane, such as the intersections of some major highways and near important transportation hubs. The center points of these clusters identify some areas where accidents frequently occur. Although the accident density is lower than that in the city center, they are still areas that need to be monitored.

Risk factor analysis: In these areas, the risk of accidents may come from sudden lane changes of high-speed vehicles, potential defects in road design (such as lack of sufficient entrances and exits or U-turn areas), unclear traffic signs, and poor lighting conditions at night.

Countermeasures and suggestions: For these areas, speed monitoring equipment can be added, road design can be optimized (such as adding isolation guardrails and clear traffic signs), and night lighting can be strengthened to reduce the probability of accidents.

Cluster shape and range

Observing each cluster in the DBSCAN clustering results, it can be found that the shape and range of each cluster are different. This difference reflects the geographical and traffic characteristics of each high-risk area. The following is a detailed interpretation of the shape and range of the main clusters:



Cluster 0 shape and coverage:

Range: Cluster 0 covers the main roads and commercial areas in and around the center of Brisbane. The cluster presents a concentrated distribution on the map, showing that the center of Brisbane is a high-incidence area of traffic accidents.

Shape analysis: The shape of cluster 0 is relatively compact and concentrated, which is consistent with the characteristics of dense accident points in the city center. This concentrated distribution may be caused by the complex road network in the city center, traffic congestion, and large pedestrian and non-motor vehicle traffic.

Traffic arterial impact: The shape of this cluster extends to major traffic arterials, especially those connected to freeways or expressways, indicating that these intersections and connecting areas are accident-prone areas and deserve further attention and monitoring.

Shape and range of other clusters (such as clusters 1 to 7):

Range: Clusters 1 to 7 are more dispersed, covering some minor transportation hubs and areas along freeways outside Brisbane. Compared with cluster 0, these clusters are wider and more irregular in shape, indicating that the distribution of accidents in these areas is more dispersed.

Shape characteristics: The shapes of these clusters are irregular and more ductile, indicating that the accident points extend along some major traffic routes rather than being concentrated in a small area. This distribution is usually seen at freeway intersections or main arterials leading into the city, with accident points distributed along the road, indicating the risk of accidents caused by high speeds and frequent lane changes.

Impact of transportation hubs: The shape and distribution characteristics of these clusters also show the influence of major transportation hubs, such as the concentration of accidents in areas such as highway-city connection points and ring roads. These areas are high-risk areas due to large traffic volume, high speed and frequent turns.

6.Limitations and Future work

Through DBSCAN cluster analysis, we can effectively identify multiple high-risk areas for traffic accidents in Queensland, and clearly divide the shapes and ranges of different clusters from the geographical

location. These findings not only help traffic management departments to formulate targeted measures, but also provide ideas for further research and improvement:

Continue to optimize the division of clusters: You can try to adjust the parameters of DBSCAN (such as neighborhood radius ϵ and minimum sample number MinPts) to optimize the clustering effect, so that the shape and range of each cluster are more in line with the actual geographical characteristics.

Combine more features for analysis: In addition to geographical location, additional features such as weather, time (such as peak and off-peak hours), road type, etc. can be introduced to further analyze the factors affecting traffic accidents and make the identification of high-risk areas more comprehensive.

Dynamic analysis and time series changes: Further research can consider adding the time dimension and conducting dynamic analysis on the accident distribution in different time periods to discover the changing patterns of high-risk areas at different times and help formulate time-based traffic management strategies. Evaluation Metrics

10.Conclusion

In this study, we successfully combined traffic crash data from 2001 to 2023 with auxiliary road roughness information from the Queensland Open Data Portal to predict the severity of traffic crashes and identify high-risk areas in Queensland. By leveraging advanced machine learning techniques, including random forests, support vector machines (SVMs), and DBSCAN, we gained a comprehensive understanding of crash severity factors and identified specific locations prone to crashes.

The results show that combining crash location data with road condition attributes can improve the accuracy of severity predictions. The SVM and random forest models provided valuable results on the likelihood of severe crashes based on environmental and contextual factors, while DBSCAN effectively highlighted the clustering of high-risk areas, especially in densely populated and heavily trafficked areas. The application of SMOTE improved the model's ability to detect minority class instances, although it brought a reduction in overall classification accuracy, indicating that there is room for further optimization in balancing techniques.

Overall, this study provides a data-driven framework that can support Queensland's traffic safety strategy, enabling policymakers to effectively allocate resources and implement targeted safety

interventions. Future studies could explore the incorporation of additional data features, dynamic time-based risk assessment, and alternative machine learning models to further improve the accuracy of predictions and the applicability of the findings.

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